

Riley Mahn

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Software engineer building agentic workflows and tooling for AI experimentation and evaluation. Combines models, scripted pipelines, and explicit controls over context and execution. Background in aerospace software verification, surgical-system quality testing, scientific instruments, and biomolecular engineering.

Python · Docker · pytest · experimental design · LLM harnesses (Claude Code, Codex, OpenCode) · Harbor / Terminal-Bench-style tasks · FastAPI · CI/CD

EXPERIENCE

AI Evaluation Engineer (contract) — Handshake AI

Apr 2026 – Present · Remote

Contract fellow building evaluations of frontier coding agents: task design, rubrics, blinded multi-model runs, trajectory grading, and the local execution platform those evals run on.

Ivy (live A/B agent evals)

- Built a Python/FastAPI operator workstation for blinded two-arm evaluations, with separate Docker harness copies to isolate model workspaces and preserve comparable execution conditions.
- Implemented verification gates that require evidence from the running workspace rather than relying on a model's self-report.
- Built a hybrid grading pipeline: independent arm evaluation, score freeze, then rationale writing, with checks for evidence consistency and cross-arm contamination. Recorded verification on 28 Aug 2026: 286 grading-tool tests and 293 cockpit tests passed.
- Authored offline benchmark tasks (Python piece-table; C++ pugixml performance) with 15–19 pass/fail rubrics and a four-model parallel runner. Calibrated for sweet-spot difficulty: at least one of four models passes and at least one fails.
- Operator status board with per-arm stage meters and stall KPIs so a stuck run is visible without staring at logs.

Dynamo, Helix, Vox, Control Room

- Design and rework Terminal-Bench-style tasks with hidden verifiers, automated grading, and Harbor validation against oracle and no-op baselines. Calibrate difficulty with repeated agent trials, distinguishing model failures from timeouts.
- Helix: golden patches and fail-to-pass tests for real GitHub issues, run in Docker (including Firecracker, Yew, NeMo Guardrails).
- Vox: 1–7 Likert grading of coding-agent communication (character, voice, structure, interaction), iterating the grading prompt against practice packs.
- Control Room: separate Codex/Claude Docker harness launcher with path isolation from the Ivy cockpit (~47 pytest cases).

Software / System Tester (contract via Brooksource) - Medtronic

Feb - May 2026

Quality testing for surgical-navigation and robotic systems, including Stealth AXIS, StealthStation S8, and Mazor, in a software/system testing role.

Software Verification / DevOps Engineer — Boeing

Apr 2022 – Nov 2024 · Denver, CO (remote)

Verification of actuator-control embedded software for military aircraft. The job was to make “this software is ready” something you could defend in a Test Readiness Review.

- Built and ran a Bamboo/Jenkins CI system that continuously executed on the order of 4,000 verification tests, with Python for statistical analysis, visualization, and test execution.
- Integrated GitLab and Jira APIs so test results, defects, and compliance evidence stayed traceable.
- Test conductor for Test Readiness Reviews and Formal Qualification Testing. Presented verification status and supporting evidence to stakeholders.
- Built a real-time motor-position monitor used to catch regressions that batch tests missed.

Lead QA Test Automation Engineer — Agilent Technologies

Dec 2019 – Apr 2022 · Santa Clara, CA (Lead Apr 2020 – Apr 2022)

- Built a C# / SpecFlow automation platform from scratch for software that drives chromatography and mass-spectrometry instruments. Scrum Master for the automation team.
- Stood up Jenkins CI with DevOps for scheduled, resource-aware test execution.
- Measured automation impact (calendar time vs. time saved) and used that to change what the team automated next.
- Configured distributed data-acquisition vs. processing so instrument runs and analysis did not share a single point of failure.

Chief Research Associate, Analytical Chemistry — i-Cultiver

Aug 2018 – Dec 2019 · Stanford, CA

Designed chemical assays with Stanford collaborators, ran ICP-OES/MS, XRF, and C/N analysis, and analyzed 500+ samples. Wrote the reports that changed what the company did next.

EDUCATION

B.S. Biomolecular Engineering, University of California, Santa Cruz, 2017

Minors: Bioinformatics; Applied Mathematics

Computational biology tooling in Python/SciPy; MATLAB statistical analysis; genomic assembly. Founded the Bioengineering Club. iGEM 2016 (metabolic pathway in *B. subtilis*). Undergraduate research in the Partch and Bowman labs.

SELECTED METHOD

- Calibrate tasks against known-good and no-op baselines, distinguishing model failures from timeouts and infrastructure errors.
- Prefer code for objective checks. Validate LLM judges against independently labeled examples before integration.
- Freeze scores before writing comparative rationales.